

Arrays

Question 1 (Easy) — Find Expensive Products

```
let prices= [100,250,500,150,700];
```

Create a new array containing only prices greater than 300.

What is this question asking?

Imagine you're building an e-commerce website.

You only want to show premium products that cost more than ₹300.

Input:

```
[100,250,500,150,700]
```

Output:

```
[500,700]
```

Concepts Tested

- Arrays
- filter()

Question 2 (Moderate) — Student Average

```
let marks= [80,90,70,85,95];
```

Calculate the average marks of all students.

What is this question asking?

A teacher has marks of students stored inside an array.

You need to:

1. Find total marks.
2. Divide by number of students.

Output:

84

Concepts Tested

- Arrays
- `reduce()`
- `length` property

Question 3 (Hard) — Most Frequent Number

```
let numbers= [1,2,3,2,4,2,5,1,1,1];
```

Find the number that appears the most.

What is this question asking?

You're given an array containing repeated values.

You need to count how many times each number appears and find the highest one.

Output:

1

Because:

```
1 appears 4 times  
2 appears 3 times
```

Concepts Tested

- Arrays
 - Loops
 - Objects for counting
 - Problem solving
-

Objects

Question 4 (Easy) — Update User Age

```
let user = {  
  name: "Ritik",  
  age: 20  
};
```

Update age to 21.

What is this question asking?

You already have an object.

Just modify one property.

Expected result:

```
{  
  name: "Ritik",  
  age: 21  
}
```

Concepts Tested

- Object property access
 - Object update
-

Question 5 (Moderate) — Print User Information

```
let user = {  
  name: "Ritik",  
  age: 20,  
  city: "Bhopal"  
};
```

Print:

```
Name: Ritik  
Age: 20  
City: Bhopal
```

using a loop.

What is this question asking?

Instead of manually writing:

```
console.log(user.name);  
console.log(user.age);
```

Loop through the object dynamically.

Concepts Tested

- Objects
- Object.entries()
- for...of

Question 6 (Hard) — Highest Paid Employee

```
let employees = {  
  aman: 25000,  
  ritik: 50000,
```

```
priya:45000  
};
```

Find the employee with the highest salary.

What is this question asking?

Compare all salaries and return:

```
Ritik
```

Concepts Tested

- Objects
- Loops
- Comparisons

Functions

Question 7 (Easy) — Greeting Function

Create a function:

```
greet(name)
```

Output:

```
Hello Ritik
```

What is this question asking?

Learn:

- Function declaration
- Parameters

- Arguments
-

Question 8 (Moderate) — Discount Calculator

Create a function:

```
calculateDiscount(price)
```

Rules:

- 10% discount

Input:

```
500
```

Output:

```
450
```

What is this question asking?

Take input.

Perform calculation.

Return result.

Concepts Tested

- Functions
 - Parameters
 - Return
-

Question 9 (Hard) — Dynamic Sum Function

Create a function:

```
sum(...numbers)
```

that can add any amount of numbers.

Example:

```
sum(1, 2, 3)
```

Output:

```
6
```

Example:

```
sum(1, 2, 3, 4, 5)
```

Output:

```
15
```

What is this question asking?

The function should not depend on fixed parameters.

It should work for unlimited inputs.

Concepts Tested

- Functions
- Rest Parameters
- `reduce()`

Arrays + Objects + Functions Together

These are the most important because real projects use all three together.

Question 10 (Easy) — Find Adult Users

```
let users = [  
  { name: "Ritik", age: 20 },  
  { name: "Aman", age: 16 },  
  { name: "Priya", age: 25 }  
];
```

Create a function:

```
getAdults(users)
```

Return only users whose age is 18 or above.

What is this question asking?

You have:

- Array
- Objects inside array
- Function

Output:

```
[  
  { name: "Ritik", age: 20 },  
  { name: "Priya", age: 25 }  
]
```

Concepts Tested

- Array of objects
- filter()
- Functions

Question 11 (Moderate) — Shopping Cart Total

```
let cart = [
  { name: "Mouse", price: 500, qty: 2 },
  { name: "Keyboard", price: 1000, qty: 1 },
  { name: "Monitor", price: 10000, qty: 1 }
];
```

Create a function:

```
getCartTotal(cart)
```

Output:

```
12000
```

What is this question asking?

Each item has:

```
price × quantity
```

Calculate total bill.

Concepts Tested

- Array of objects
- Functions
- `reduce()`

Question 12 (Hard) — Student Grade Report

```
let students = [
  {
    name: "Ritik",
```

```
    marks: [80, 90, 85]
  },
  {
    name: "Aman",
    marks: [50, 40, 60]
  }
];
```

Create a function:

```
generateReport(students)
```

Return:

```
[
  {
    name: "Ritik",
    average: 85,
    grade: "A"
  },
  {
    name: "Aman",
    average: 50,
    grade: "C"
  }
]
```

What is this question asking?

For every student:

1. Calculate average.
2. Decide grade.
3. Create a new object.
4. Return a new array.

This is extremely close to real-world frontend interview questions.

Concepts Tested

- Arrays
 - Nested Arrays
 - Objects
 - Functions
 - map()
 - reduce()
 - Conditional Logic
-

Final Challenge (Very Hard)

Mini Library Management System

Create:

```
addBook(title,author)
borrowBook(id)
returnBook(id)
showAvailableBooks()
```

Books should be stored as:

```
{
  id:1,
  title:"Atomic Habits",
  author:"James Clear",
  borrowed:false
}
```

What is this question asking?

You need to manage a collection of books.

Tasks:

1. Add new books.

2. Borrow books.
3. Return books.
4. Display available books.